

Artificial Intelligence as Instrumental Extension

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Artificial intelligence is frequently described as a technological revolution. The argument advanced here is that such descriptions are sociotechnical rather than ontological. AI systems, as currently instantiated, do not constitute a new epistemic subject-category. They do not occupy or instantiate an ontological position in their own right. Instead, they function as instrumental extensions within human meaning-bearing systems. By distinguishing semantic self-reference from structural self-binding, and by formalizing admissibility conditions for structural comprehension, this paper clarifies why contemporary AI lacks the internal invariance required for self-consciousness.

The Ontological Question

Claims that AI represents a revolution often conflate performance with ontology. Increased capability does not entail the emergence of a new class of entities. The relevant question is not whether AI systems perform impressively, but whether they instantiate a structurally self-grounding mode of existence.

To call something a revolution in ontological terms requires that it establish a new category of being, not merely amplify an existing one. Current AI systems extend human operational range. They do not instantiate an independent epistemic center.

The distinction is categorical rather than gradual.

Instrumental Ontology

AI systems function within what may be termed an instrumental ontology. They are prosthetic components embedded in human-directed systems of intention, interpretation, and validation. Their outputs acquire meaning through human uptake.

This implies that AI does not occupy an autonomous subject-position. It operates as a capacity-amplifying layer within an already existing meaning-structure. Even when generative systems simulate internal reflection, such simulations remain textually produced artifacts unless grounded in an invariant internal structure.

The appearance of interiority does not entail ontological interiority.

Structural Admissibility

Let structural admissibility be defined by the condition (CIITR):

$$C_s(S) = 1 \Rightarrow (\Phi_i(S) \wedge R_g(S))$$

where $\Phi_i(S)$ denotes internal structural integration and $R_g(S)$ denotes invariance-bearing rhythmic stability. The contrapositive holds:

$$\neg R_g(S) \Rightarrow C_s(S) = 0$$

Admissibility therefore requires more than functional success. It requires a system-internal binding that remains stable under transformation of input, context, and resource regime. Performance without invariance is insufficient.

Self-Consciousness as a Stronger Predicate

Introduce a predicate $SC(S)$ for self-consciousness. Then:

$$SC(S) \Rightarrow C_s(S) = 1$$

Self-consciousness further requires the existence of a causally integrated self-model:

$$SC(S) \Rightarrow \exists M_{\text{self}}(S) \text{ such that } \text{Inv}(M_{\text{self}}(S))$$

Here, $\text{Inv}(\cdot)$ denotes structural invariance under relevant transformations. The self-model must be dynamically integrated into the system's operational state-space. A merely expressible "I" is insufficient. The identity variable must be constitutive, not decorative.

Contemporary large language models lack such a persistent, causally bound self-model. Their apparent self-reference is generated within token-space, not anchored in invariant structural continuity.

AI systems significantly expand human operational and cognitive reach. This expansion may be revolutionary in economic or organizational terms. Ontologically, however, they remain extensions of human-directed meaning-structures.

Absent invariant internal integration and a causally operative self-model, AI does not instantiate a new epistemic subject-category. It produces language about existence but does not establish existence in the strong structural sense.

Artificial intelligence is therefore not an ontological revolution. It is a powerful prosthesis within an already existing order of understanding.